

Total No. of Questions : 10]

SEAT No. :

PE-5954

[Total No. of Pages : 2

[6583]-330

T.E. (A.I. & M.L.) (Honours)

COMPUTATIONAL STATISTICS

(2019 Pattern) (Semester - V) (310301)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3. or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Make suitable assumptions whenever necessary.

Q1) a) What is Mean, Median and Mode with example? [5]

b) Difference between Univariate and Bivariate Analysis? [5]

OR

Q2) a) Explain the Histogram, Statistical Thinking with example? [5]

b) Difference between the Standard Deviation & Harmonic Mean with example? [5]

Q3) a) Explain Normal Distributions with example? [5]

b) Difference between the Characterizing a distribution & Discrete Distributions? [5]

OR

Q4) a) Write down short note on Lognormal, Weigh bull, Exponential, Uniform? [5]

b) What is Poisson Distribution with suitable example? [5]

P.T.O.

- Q5)** a) Explain the Hypothesis Concept with example. [8]
b) What is Confusion Matrix in detail. [8]

OR

- Q6)** a) What is Test on Numerical Data. Distribution of a Sample Mean with example. [8]
b) Comparison of Two Groups Multiple Groups. [8]

- Q7)** a) Write down the short no Normalization - Feature Scaling. [8]
b) Difference between the Ridge Regression, Lasso Regression. [8]

OR

- Q8)** a) Explain in detail Cross Validation Techniques-K-fold with example. [8]
b) Difference between the Grid Search CV & CV Error. [8]

- Q9)** a) What is the different Dimensionality Reduction Techniques Explain with example. [9]
b) Where we use the Chi-square method with example. [9]

OR

- Q10)** a) When we used the Correlation coefficient, Rank Correlation & how with example. [9]
b) Difference between the Bayes' Theorem & Monte Carlo Method. [9]
